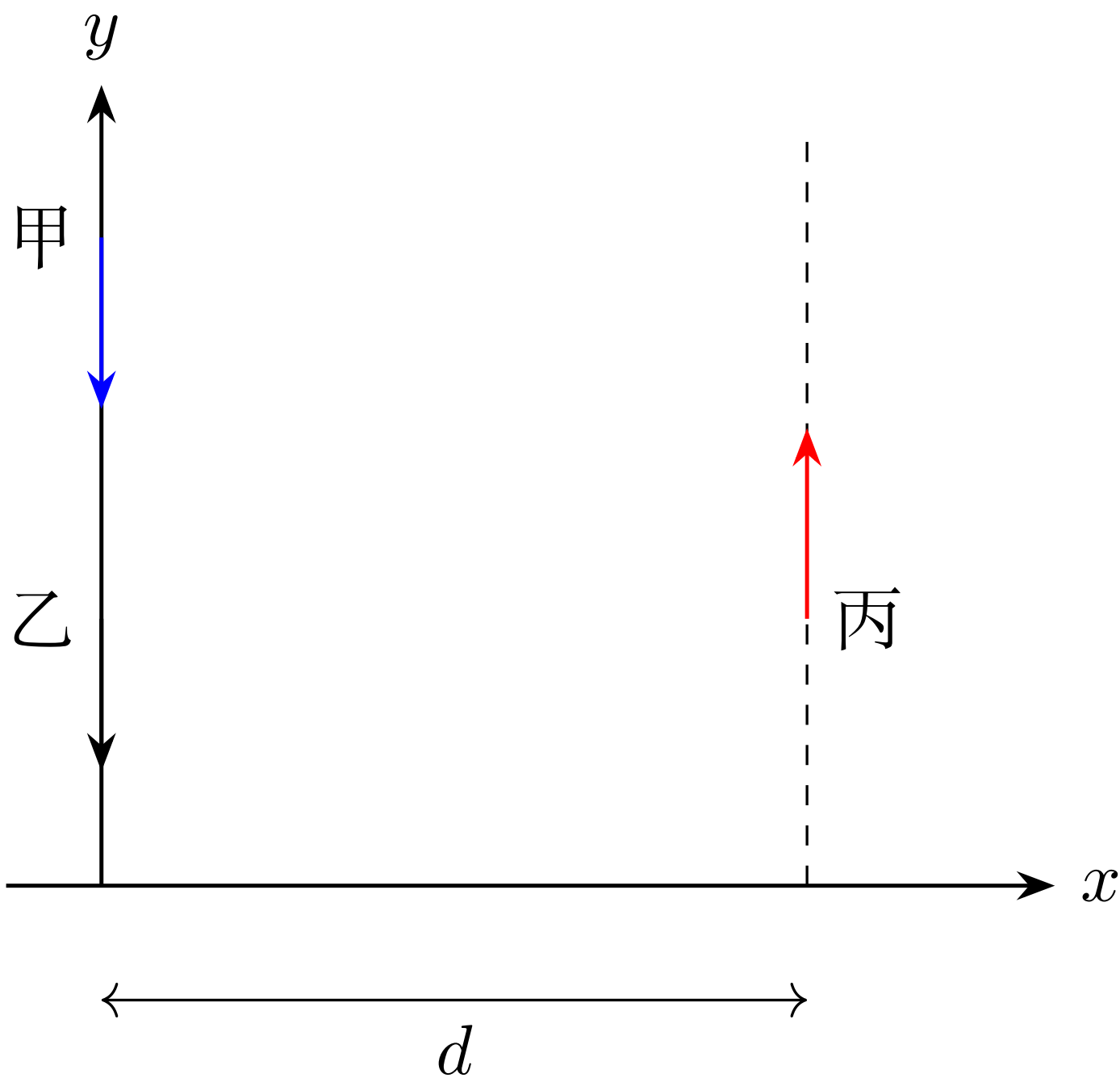$v_{甲} = -2c/3,\ v_{乙} = -c/3,\ v_{丙} = 2c/3$